

Reading Support: Embedding and Homomorphisms

Companion to Week 11

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Embedding places one structured object within another

Supports: Syntactic embedding is structural containment

Syntax allows an expression to occupy a position inside another structured expression. A relative clause can occur inside a noun phrase, and a complement clause can occur as an argument of a verb. The containing expression may itself occupy such a position again.

Embedding concerns the structural relation. A long string without containment and a short clause containing another clause must be distinguished.

MOLR §6.1

Semantic composition requires admissible functional connection

Supports: Semantic embedding composes functions and arguments

Semantic values combine when one value is defined for an argument of the kind supplied by the other. A type records this admissibility condition: the argument type corresponds to what the lambda binds, and the result type to the value of its body.

The notation does not explain the meaning. It records which compositions the chosen analysis permits.

MOLR §6.1

A homomorphism preserves a specified operation

Supports: A homomorphism preserves an operation

For structures with operations $\star_A$ and $\star_B$, a mapping h is homomorphic when $h(x \star_A y) = h(x) \star_B h(y)$. The elements and operations in the two domains need not be identical.

The claim has content only after we identify which operation and which structural relation the mapping preserves.

MOLR §6.2

Possible homomorphisms exceed the mappings found in a language

Supports: Linguistic homomorphisms are contingent

Mathematics permits many structure-preserving mappings between possible syntactic and semantic objects. A language selects only some through its lexicon, categories, constructions, and practices of interpretation.

These mappings are contingent because another internally coherent language may realize the dependency differently. Formal possibility cannot choose among the empirical alternatives.

MOLR §6.2

Admissibility operates repeatedly across a derivation

Supports: Admissibility filters possible form-meaning pairings

At each stage, composition produces a possible object only when the domains connect. Grammatical categories and lexical values further restrict which possible composition belongs to the language. Context can then admit or reject a use of the resulting expression.

The process is a cycle of composition followed by admissibility, not a single initial filter that generates an unrestricted space of variants.

MOLR §6.2

Lexical event structure affects semantic combination

Supports: Aspect shows why lexical correspondence matters

The contrast between win and play under duration and frame adverbials follows from the event structures assigned to the predicates. A modifier that composes directly with one value may require coercion or a richer construction with another.

The admissibility of the whole therefore depends on the lexical mapping carried into composition.

MOLR §6.2

Contingency connects acquisition with diversity

Supports: Contingency links the three fundamental questions

Children acquire correspondences sustained by the linguistic activity around them. Continuity concerns how early correspondences develop into adult systems. Diversity follows because different communities can sustain different form-meaning mappings while preserving comparable dependencies.

The task is to explain convergence without replacing empirical variation by predetermined formal switches.

MOLR §6.3

Decomposition does not entail embedding

Supports: Morphology and phonology can decompose without embedding

Morphological and phonological objects have internal parts and dependencies. This does not by itself mean that an object of the same structural kind occupies a position inside another one.

Regular and linear relations can characterize substantial internal organization. Recursion should be attributed only when the continuing relation actually re-enters its own domain.

MOLR §6.4

Turkish compounds require morphological constituency

Supports: Turkish compounds show structured decomposition

Turkish compound marking relates the possessive suffix to a compound constituent rather than to any adjacent sequence that happens to precede it. Alternative bracketings predict different units for interpretation and further composition.

The analysis must preserve these morphological dependencies without assuming that every nested-looking sequence constitutes syntactic embedding.

MOLR §6.4